Crawler and listing inventory: session report

Worktree **crawler-inventory**, branch **feat/crawler-inventory**, 2026-10-17. Plan: doc/crawler-inventory-plan.md (approved 2026-10-17). Reference: doc/crawled-listings.md sections 4, 4a, 4b, 5, 9 and 11.

1. Summary

The 2024 scrapy spiders were dead against today's dba.dk and tradera.com. This session moved the crawler into the repository (**crawler/**), rewrote both spiders for the current markup, made them emit the importer's own flat shape (schema **crawled-item/2**), and gave the listing sub-context what it needs to tell a usable listing from a poor one: a **quality grade** (GOOD / WEAK / REJECT with issue codes) computed at import, **index gating** by grade, a bounded **liveness check** (HEAD on url and image, 404/410 = GONE at once, dead image = downgrade), and an **inventory endpoint** for the admin edge. A real run of both spiders imported 639 listings into the live database and the OCS index with nothing rejected.

2. What shipped

Area	Change
Schema (db-reactive)	crawled_listing: quality_grade, quality_issues (JSON), url_checked_at, url_ok, image_ok, key (status, url_checked_at). New table listing_import_run (one row per imported file with the ImportReport counts). Applied to the live announce_reactive_db; announce_table.sql + announce_reactive_db.sql regenerated; mappers regenerated (restricted generator config) and reactified; hand-written statements upsert (resets image_ok when the image url changes), selectForLivenessCheck, updateLiveness, updateQuality, selectInventoryCounts.
Domain	ListingQuality (grade + issues; hard issues no_url / no_title / no_image reject, soft issues url_pattern, image_not_https, image_host_unknown, no_price, no_published_at, no_brief, no_attributes, image_dead weaken), ListingGrade, LinkChecker port + LinkStatus, ListingImportRun(+Repository), InventoryCount; ListingSource knows each site's canonical item-url pattern and image hosts; CrawledListing carries quality and liveness columns, seenAgain() regrades and forgets the image verdict when the image changed, checked() applies a probe result.
Application	IndexGating (announce.listing.index-grades, default GOOD,WEAK); ListingImportService grades, gates, records the run, retries a failed index call twice and counts indexFailed; ListingReindexService grades ungraded rows on the way and pushes passing rows only; ListingLivenessService + LivenessReport; ListingInventoryService + ListingInventory.
Infrastructure	CrawledItemV2Parser (generic v2 lines, used by DbaFeedParser / TraderaFeedParser when a line carries the schema marker); WebClientLinkChecker (HEAD, GET Range fallback on 405, redirects, 10 s timeout, bot user agent); ListingLivenessJob (announce.listing.liveness.cron, every 15 min, batch 100, 1 s delay); MyBatis adapters for the new statements and table.
REST	GET /api/admin/listings/inventory (per source: active, gone, grades, missingImage, deadLink, checked, lastImport); POST /api/admin/listings/liveness (one batch now). The admin edge already routes /api/admin/listings/**.
Crawler (crawler/)	Package announce_crawler: settings (robots obeyed, 1 request at a time per site, 1 s delay + autothrottle, honest user agent Mozilla/5.0 (compatible; AnnounceBot/0.1; +https://github.com/joviste/announce)), pipeline (schema stamp, crawledAt, 255-char brief, drop without url/title), pure parsers (parsing/dba.py, parsing/tradera.py, parsing/text.py), spiders dba_cars and tradera_cars (-a max_pages, max_items, detail, start_page, registration_class / category), feed _.jsonl.part renamed by run.sh, pinned requirements.txt, pytest suite over trimmed live-page fixtures.
Config	announce-management application.yml: listing.index-grades, listing.liveness.*, resilience4j.timelimiter.instances.ocs-indexer 15 s.
Docs	doc/crawled-listings.md (v2 feed, quality rules, liveness, inventory, runbook); doc/crawler-inventory-plan.md status + deviations; this report.

3. Survey facts of 2026-10-17 (fresh fetch, confirms the 2026-09-29 survey)

Site	What the spider reads today
dba.dk list	/mobility/search/car?registration_class=1&page=N: JSON-LD CollectionPage in script#seoStructuredData with 48 Products (name, url, image on images.dbastatic.dk, brand, model, offers.price DKK, itemCondition); an article.sf-search-ad per product with the variant line, '2015 ■ 223.000 km ■ Diesel ■ Manuelt', the city, 'Privat ■ Service' and the age ('2 t.', '3 dage'). About 18,800 cars in class 1.
dba.dk item	JSON-LD Product with the image list; label/value rows (Modelår, Kilometertal, Geartype, Drivmiddel; dt/dd 1. registrering; p/p Annonce-ID, Opdateret); 'Placering' with '8520 Lystrup'. No creation date.
tradera.com list	/category/1001?paging=N: 80 div[data-item-card-id] cards (data-item-type ContactOnly / Auction / AuctionBin / PureBin), title + href on a[data-testid=item-card-image], picture source set on img.tradera.net (medium-fit used), [data-testid=price] '374 800 kr', attribute spans (year, '9 347 mil', fuel, gearbox), 'Inkom 3 h sedan' or, on auctions, 'Sluttid 23 okt 17:59'.
tradera.com item	React server payload (self.__next_f.push) with escaped itemDetails: startDate, endDate, itemType, fixedPrice, openingBid. JSON-LD Product with the full description (not stored).
robots / UA	Both robots.txt allow the list and item pages; both sites answered 200 to the bot user agent, so no disguise was needed.

4. Measurements

Run	Result
Spiders, 5 pages each	239 dba items (pages of 48/45/49/48/49 JSON-LD products), 400 tradera items (80 a page), no item dropped, a few seconds per spider.
Import (polling flow, live DB + OCS)	dba: 239 seen, 238 inserted + 1 stored without document (first bulk call to the cold indexer tripped the 1 s time limit; now retried and counted as indexFailed), 238 GOOD. tradera: 400 inserted, 365 GOOD, 35 WEAK (auction cards show Sluttid instead of Inkom, so no published date), 0 REJECT, 0 rejected.
Inventory	DBA active 293 (239 GOOD + 54 ungraded 2024 rows), missingImage 4; TRADERA active 400. lastImport file and counts shown per source.
Liveness batch (30 rows, real HEAD)	21 of the 2024 legacy rows GONE (dba answers 404 for the old /slug/id-<digits>/ urls) and removed from the index; 9 rows url_ok=1, image_ok=1. Today's urls and images answer 200.
Search tier	searcher: q=peugeot&source=dba 25 hits, q=volvo&source=tradera 63 hits, documents carry source, external_url, source_category, km, model_year, fuel, gearbox.

5. Tests

mvn -pl db-reactive test: 75 tests, 0 failures (BUILD SUCCESS, 2026-10-18). CrawledListingMapperTest grew to 5 tests (liveness + quality columns, upsert keeps the verdict for the same image and resets it for a new one, inventory counts, import-run rows).

mvn -pl announce-management -P '!webapp' test: 71 tests, 0 failures (BUILD SUCCESS, 2026-10-18). ListingImportEndToEndTest has 12 ordered checks: the five-row fixture (3 GOOD, 2 WEAK, 5 index calls, then none), the inventory for it, a price change, the sweep, /out, the GONE-then-seen row, the admin endpoints, the polling flow, a v2 item without image stored REJECT and never indexed (a reindex skips it), the liveness check with a fake LinkChecker (410 url = GONE + removed + /out 410; dead image = GOOD to WEAK with image_dead and a reindex; live again = GOOD), the 2024 dba_cars.json (54 rows, 4 REJECT) and the newest real v2 feeds of both spiders (nothing rejected, at least 90 % GOOD, no index failure). New unit tests: ListingQualityTest (5), FeedParsersTest v2 cases (2).

pytest in crawler/: 14 tests over trimmed copies of today's pages (every field of both list parsers, the detail passes, paging and caps, the pipeline rules, the text helpers).

6. Deviations from the approved plan and gotchas

• Package renamed to announce_crawler; venv on anaconda's Python 3.9 because the pyenv 3.11 and 3.12 builds lack the bz2 module that scrapy imports at startup (ModuleNotFoundError: _bz2).

• listing_import_run is a new table; the plan only asked for 'last import time and file'.

• ImportReport gained indexFailed; a failed index call is retried twice and never counted as rejected (the row is stored, the reindex catches up).

• Any number of soft issues gives WEAK; url pattern, image scheme/host and a missing fixed price are soft, so the legacy 2024 rows stay indexable.

• The mybatis generator plugin ignores -Dmybatis.generator.configurationFile when the pom pins the file: swap a restricted config in and out of mybatis-generator/generatorConfig.xml, run from db-reactive/, then reactify.

• A running announce-management picks up any dba_*.json dropped into the import directory within a minute (it moved dba_cars.json to done/ during a test run); stop the service before running the end-to-end suite with the real file.

• The legacy dba fixtures share one image url, so a fake link verdict keyed by url hits all three rows; the liveness test uses the v2 Peugeot row instead.

• Scrapy 2.13 deprecates REQUEST_FINGERPRINTER_IMPLEMENTATION and start_requests(); the spiders use async start().

7. Runbook (short)

```
cd crawler && python3 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python -m pytest
./run.sh # MAX_PAGES=5 DETAIL=0 by default, feeds into ~/announce_crawled_data
MAX_PAGES=20 DETAIL=1 ./run.sh
curl localhost:8085/api/admin/listings/inventory
curl -X POST localhost:8085/api/admin/listings/liveness
curl -X POST localhost:8085/api/admin/listings/reindex # also grades rows imported before
2026-10-17
```

8. Next steps

• Commit (not done: the repo rule is commit on request), merge into announce-jo-branch, then replace ~/Desktop/python_projects/scrapy_projects/tutorial by a symlink to crawler/.

• Run the reindex once on the live service so the 54 rows of the 2024 feed get a grade (four will become REJECT and leave the index; the liveness job is removing the dead ones anyway).

• Admin SPA: a screen on /api/admin/listings/inventory (out of scope here).

• Verify dba registration_class 2 and 3 before crawling them; add a cron or systemd timer for run.sh; Kafka transport later.